

Simulated Peer Review — Round 3 (two-decoy dense benchmark):
A Lineshape Correlation Coefficient for 2D HSQC Similarity

Editorial simulation (4 referees + adversarial verification + handling editor)

Simulated peer review — Round 3

Manuscript: *A Lineshape Correlation Coefficient for Robust Similarity of Two-Dimensional HSQC NMR Spectra* (D. Lee, KBSI), re-reviewed after the addition of an **expanded dense ^1H – ^{15}N benchmark with two decoy proteins** (main-text Table 2; SI “Expanded benchmark — a second decoy protein”, Tables S8–S9). Method: four independent referees, adversarial verification of every MAJOR against the actual files, then a handling-editor re-decision. This round specifically asks whether the new data changes the Round-2 disposition (reject-and-transfer).

Round-3 decision: REJECT (reject-and-transfer) — unchanged outcome, one round-2 objection now closed

Referee	Lens	Round 2	Round 3
Referee 1	Novelty, significance, journal fit	Reject	Reject
Referee 2	Technical correctness & statistics	Major revision	Reject-and-transfer
Referee 3	Benchmarking fairness & reproducibility	Major revision	Major revision
Referee 4	Presentation & format	Reject	Reject
Handling editor	Consolidated	Reject-and-transfer	Reject-and-transfer

Verification outcome: of 14 MAJOR findings raised this round, adversarial verification CONFIRMED 5 and judged the rest OVERSTATED/REFUTED. Crucially, **three of the five confirmed findings were newly introduced by this revision** and are fixable defects (now corrected — see below); the other two are the standing significance/reproducibility items.

What the new data changed (verified)

The two-decoy benchmark is a **genuine, reproducible strengthening** that closes one round-2 item:

- **Round-2 cap “the mean-centring novelty and the dense margin rest on a single negative spectrum (n=1)” is materially resolved.** The negative side is now a distribution over **two distinct folds** — OAA (277 K) and the kinase domain EphB3 (900 MHz) — scored against an **independent 23-point PRL3 apo→olsalazine titration**. LCC keeps both decoys below *all 23* same-protein points (worst same 0.82 > worst decoy 0.41; margin +0.41), mean-centring still earns the gap against a real negative distribution (LCC separation 0.57 vs un-centred cosine 0.46), and the titration decays smoothly 0.98→0.82 — a clean demonstration of the graded shift tolerance on real data.
- **Referees 2 and 3 independently re-ran bench_nhsqc.py:** Table 2 and Tables S8/S9 reproduce to the digit, the separation/margin definitions are correct and reduce to the Table-1 definitions at one decoy, the reference is genuinely apo, and the decoys carry real overlapping amide structure (tree/NN saturate at 0.75–0.96) that LCC nonetheless separates — so it is not an empty-window artefact.

Why the outcome is unchanged. Round 2 had *already* down-ranked the single-negative concern to MINOR; its **decisive** gate was *broad chemical significance for a general journal*, closeable only by “at least one real chemical result where LCC changes a conclusion an incumbent gets wrong.” Table 2 adds more same/different **discrimination scores** (23 comparisons + 2 decoys) — the same evidence type at larger n — not an application: no binder identified, no CSP site mapped, no unknown dereplicated, no QC/comparability failure caught. The HOS framing (refs 11–16) is well-cited motivation, not a demonstrated comparability result. So rigor is up (helping the specialist-venue case) but the general-journal gate is untouched.

Confirmed findings (survived adversarial verification)

Newly introduced by this revision — now fixed:

1. [MAJOR, fixed] **Factual overclaim “the only margin.”** Main text called LCC’s the “only margin that pushes both decoy proteins below every same-protein score,” yet all six methods have positive Table S8 margins (Bin 0.25, Bin+45 0.25, tree 0.12, NN 0.02, cosine 0.35, LCC 0.41) and the same sentence cites the positive tree/NN margins. **Fixed:** reworded to “the widest margin (0.41)... whereas the tree and nearest-neighbour margins are a knife-edge (0.12 and 0.02).”
2. [MAJOR, fixed] **Reproducibility regression in bench_nhsqc.py.** The new harness hard-coded the private Nhsqc/ paths with no flag/env interface and a raw `FileNotFoundError`, undoing the round-2 `--prl3/--oaa + $PRL3_DIR` fix. **Fixed:** added `--titration/--oaa/--ephb3` flags and `$NHSQC_*` env overrides with a clean `p.error()` when the data are absent (numbers unchanged).
3. [MAJOR, fixed] **SI §S13 de-synchronised.** §S13 still called the dense margin “a single negative spectrum (§S8)” — contradicting §S8, where the second decoy was added. **Fixed:** §S13 now names the two-decoy expansion and the 0.41 margin.

Standing items (decisive / editorial — not fixable by editing):

4. [MAJOR $\rightarrow$ decisive] **No application-grade chemical result.** The reject-driving gate; unchanged since round 2. Needs new experimental chemistry, not more discrimination numbers.
 5. [MAJOR $\rightarrow$ minor at a methods venue] **The decisive dense evidence is not yet public.** `bench_nhsqc.py`, the `Nhsqc/` raw data and `results/nhsqc_dense.json` are untracked in git and the deposition is “in preparation” (no DOI), so Table 2’s “reproducible” claim is false on a public checkout. Fix by committing the harness + depositing the matrices with a DOI.
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Remaining blockers (ranked)

1. **Chemical significance for a general journal** — add at least one application where LCC corrects an incumbent’s chemical conclusion (a blind BMRB/HMDB dereplication hit at library scale under a rejection threshold, or a CSP/biosimilar-HOS screen where bin/tree/NN misrank and LCC ranks correctly). New application-grade work $\rightarrow$ resubmit, not held revision.
 2. **Deposit the dense data** (both single- and two-decoy) with a DOI; commit `bench_nhsqc.py` so the in-repo claim is true on a clean checkout.
 3. **Carryover minors** (unchanged from round 2): unlabelled $3\times$ (margin) vs $2.6\times$ (separation) multiplier; the per-regime grid step as a de-facto third parameter despite “set by spectroscopy”; the retrieval null (top-1 = 1.00 tie) as the only task-level test.
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Bottom line

The two-decoy benchmark is a **real success on the axis round 2 asked about**: the dense result no longer rests on a single negative spectrum, mean-centring’s benefit is demonstrated against two distinct folds and a 23-point real titration, and everything reproduces locally to the digit. It is **not** a success on the axis that decides venue: no chemical result was added, so the general-chemistry significance gate is untouched, and the revision briefly introduced three fixable defects (all now corrected). The intellectually honest disposition is unchanged and, if anything, sharper: **transfer to a specialized methods journal** (J. Magn. Reson., Anal. Chem., or J. Cheminform.), where this is now an even stronger, near-acceptable methods paper, with a path back to Angewandte only on a genuine, at-scale chemical demonstration.